Smart Hospital Operations, Emergency & Resource Management System

Project Duration: 2 Months

Project Type: Full-Stack Software Development

Technology Stack

- **Backend:** Python + Flask
- **Database:** MySQL
- **Frontend:** HTML5, CSS3, JavaScript, Bootstrap
- **Data Processing:** Pandas, NumPy
- **Charts:** Chart.js
- **Reports:** PDF + Excel/CSV
- **API:** REST API
- **Version Control:** Git + GitHub

1. Patient Management

The system should provide:

- Patient registration
- Unique Patient ID generation
- Patient profile
- Medical history
- Previous visit history
- Emergency registration
- Patient search and filtering

2. Appointment Management

- Doctor availability
- Appointment booking
- Rescheduling and cancellation
- Doctor-wise schedules
- Department-wise appointments
- Waiting-list management
- Appointment status tracking

3. Smart Bed Allocation

The system should automatically identify a suitable available bed based on:

- Ward type
- Bed type
- Patient gender
- Emergency priority
- ICU requirement
- Isolation requirement
- Bed availability

Example workflow:

```
Emergency Patient
        ↓
Check Requirement
        ↓
Find Available Bed
        ↓
Validate Bed Type
        ↓
Allocate Bed
        ↓
Update Bed Status
```

4. Emergency & Triage Module

For emergency patients, record:

- Blood pressure
- Heart rate
- Oxygen level
- Temperature
- Symptoms
- Emergency condition

The system should calculate an **emergency priority level** and place patients into an emergency queue.

```
Patient Registration
         ↓
Vital Signs
         ↓
Priority Calculation
         ↓
Emergency Queue
         ↓
```

```
Doctor Assignment
        ↓
Bed Allocation
```

5. Doctor Management

Manage:

- Doctor profile
- Department
- Specialization
- Available timings
- Appointments
- Current patients
- Emergency duty schedule

6. Laboratory Management

Doctor can prescribe laboratory tests.

Example:

- CBC
- Blood Sugar
- Lipid Profile
- LFT
- KFT

Workflow:

```
Doctor Prescribes Test
        ↓
Lab Test Assigned
        ↓
Sample Collected
        ↓
Test Processing
        ↓
Result Entered
        ↓
Doctor Reviews Result
```

7. Pharmacy Management

Manage:

- Medicine master

- Medicine stock
- Batch number
- Expiry date
- Purchase quantity
- Available quantity
- Prescription
- Medicine issue

The system should generate **low-stock alerts** and prevent issuing expired medicines.

8. Billing Management

Automatically calculate:

```
Consultation Charges
+
Bed/Room Charges
+
Laboratory Charges
+
Medicine Charges
+
Other Services
----------------------
Final Bill
```

Include:

- Discounts
- Taxes
- Payment status
- Multiple payment methods
- Partial payments
- Outstanding amount

9. Discharge Management

When a patient is discharged, generate a complete discharge summary containing:

- Patient details
- Diagnosis
- Doctor details
- Treatment summary
- Laboratory reports
- Medicines

- Doctor instructions
- Follow-up date
- Final bill

Generate it as a **PDF**.

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10. Role-Based Access Control

Create separate access for:

Admin

Full system access.

Receptionist

Patients + appointments + admissions.

Doctor

Patients + appointments + prescriptions + lab reports.

Nurse

Patient care + bed allocation + vital signs.

Lab Technician

Tests + samples + results.

Pharmacist

Prescriptions + medicine inventory.

Billing Staff

Bills + payments + receipts.

Each role must have a different dashboard and permissions.

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11. Advanced SQL Requirements

Students should implement:

- INNER JOIN
- LEFT JOIN
- Multiple JOINs
- GROUP BY
- HAVING
- Subqueries
- CTEs

- Window Functions
- Views
- Indexes
- Transactions
- Stored Procedures
- Triggers

Example:

Find doctors who have handled more than 50 patients in a month.

Find departments with the highest patient load.

Calculate monthly hospital revenue.

Find medicines that will expire within 30 days.

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12. Python Advanced Logic

Implement Python services for:

Automatic Bed Allocation

```
Patient Requirement
        ↓
Search Available Beds
        ↓
Apply Rules
        ↓
Select Suitable Bed
        ↓
Allocate Bed
```

Automatic Billing

```
Services
   ↓
Charges Calculation
   ↓
Discount
   ↓
Tax
   ↓
Final Amount
```

Emergency Queue

Patients should be sorted according to their calculated emergency priority.

Report Generation

Use Pandas to generate:

- Daily reports
- Weekly reports
- Monthly reports
- Department reports
- Doctor performance reports
- Revenue reports
- Pharmacy reports

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13. Admin Dashboard

Display:

```
Total Patients
Today's Appointments
Available Beds
Occupied Beds
Emergency Patients
Pending Lab Tests
Low Stock Medicines
Today's Revenue
```

Charts:

- Patient Registration Trend
- Department-wise Patients
- Bed Occupancy
- Emergency Cases
- Monthly Revenue
- Laboratory Test Statistics
- Medicine Consumption

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14. Notification System

Create notifications for:

- New appointment
- Appointment cancellation
- Emergency patient

- Lab result available
- Low medicine stock
- Medicine expiry
- Payment pending
- Patient discharge

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15. Audit Log

Every important action must be recorded.

Example:

```
10:15 AM
Receptionist created Patient P10025

10:20 AM
Doctor assigned to Patient P10025

10:45 AM
Lab Test CBC ordered

11:30 AM
CBC result uploaded

12:00 PM
Prescription created

04:00 PM
Patient discharged
```

This makes the project significantly more advanced than a normal CRUD application.

Database Tables

A possible database structure:

```
users
roles
patients
doctors
departments
appointments
emergency_cases
triage_records
```

```
wards
beds
admissions
lab_tests
lab_results
medicines
medicine_stock
prescriptions
prescription_items
```

2-Month Development Plan

Week	Work
Week 1	Requirement analysis, ER diagram, database design, UI wireframes
Week 2	Login, registration, roles, authentication, patient management
Week 3	Doctor, appointment, admission and bed

	managem ent
Week 4	Emergenc y, triage, priority calculatio n and smart bed allocation
Week 5	Laborator y and pharmacy modules
Week 6	Billing, payments and discharge managem ent
Week 7	Dashboar ds, analytics, notificatio ns, audit logs and reports
Week 8	Testing, security, optimizati on, deployme nt and

	document ation

Final Project Title

“Smart Hospital Operations, Emergency & Resource Management System”

Minimum Internship Deliverables

1. Requirement Specification Document
2. ER Diagram
3. Database Schema
4. UI Design
5. Python/Flask Application
6. MySQL Database
7. REST APIs
8. Role-Based Authentication
9. Admin Dashboard
10. Reports and PDF Generation
11. Testing Document
12. GitHub Repository
13. Final Project Report
14. Project Presentation
15. Live Demonstration